

KSHAMA R BHATT

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EDUCATION

University of Pennsylvania

Master of Science in Engineering, Computer and Information Science (MSE-CIS)

Aug 2026 – May 2028 (Expected)

Philadelphia, PA

B.M.S. College of Engineering

B.E. in Information Science and Engineering; **GPA: 9.34/10 (4.0/4.0)**

Aug 2020 – Jun 2024

Bengaluru, India

EXPERIENCE

Site Reliability Engineer, IBM

IBM Enterprise Application Service for Java (EASeJ) & IBM MQ on Cloud (MCSP)

Aug 2024 – Present

Bengaluru, India

- Operate production **OpenShift/ROKS clusters** on AWS and IBM Cloud across dev, staging, and prod; own version upgrades, worker-node scaling, and TLS reconciliation; tuned Prometheus/Grafana/Instana stack to cut unplanned downtime **30%** and MTTR **40%**.
- Lead **Sev1/Sev2 incident response**: triage, mitigation, customer comms, and structured **Root Cause Analysis (RCA)** with engineering for permanent fixes.
- Shipped end-to-end **AWS Cost Optimization Platform**: Terraform-deployed auto-tagging Lambda across 3 accounts, CUR→S3→Lambda ETL→DynamoDB, IAM-auth **API Gateway** cost-query API, and **Grafana** dashboards with Turbonomic for optimization actions.
- Own **BCDR testing** and **environment lifecycle automation**: provision and deprovision short-lived PoC environments via Terraform + Ansible, validate failover drills, verify backup restore paths.
- Built **Jenkins CI/CD** and **GitOps (ArgoCD)** workflows plus Ansible/Python automation — **Slack-native JIRA alerting**, automated **Salesforce Change Requests**, cluster validation, and config sync; drove **SOC2** audits on backup alerting, IAM, and Secrets Manager rotation.

Software Engineer Intern, IBM

IBM Watsonx & Offering Cost Calculator

Jan 2024 – Jul 2024

Bengaluru, India

- Built a **Python + React** tool integrating **WatsonxLLM** to automate manual service-management workflows across the Watsonx offering lifecycle; tuned **DB2** queries and backend logic to keep a high-traffic internal API responsive under load.

Software Engineer Intern, Hewlett Packard Enterprise

Open Source Component Selector

Feb 2023 – May 2023

Remote

- Built a full-stack tool (**Flask, MySQL, React**) letting engineers search **500+ open-source components** and filter by license compatibility, replacing manual license vetting; designed the metadata-search and filtering schema behind it.

PROJECTS & RESEARCH

Pre/Post-Deployment SRE Analysis Platform | IBM Bob, MCP, Prometheus, Loki

2nd Place, IBM Bobathon 2026

- Built an **agentic AI orchestration platform** (team Autonomous Reliability Defenders; presented at **WIST**): a master agent coordinates specialized workers (security, capacity, logs, deployment) for pre-deploy validation and post-deploy incident analysis, cutting hours of manual investigation to minutes.
- Wrote custom **Model Context Protocol (MCP) servers** for Loki and Prometheus; integrated ArgoCD, Instana, and GitHub MCPs; solved **200K context-window limits** via task decomposition and parallel agent execution.

KubePilot – Kubernetes Reliability Platform | Go, Kubernetes, Claude API, Prometheus, Grafana, OTel, Helm, Next.js

[GitHub](#)

- **8 deterministic Go analyzers** (cluster health, workloads, resources, reliability, upgrade readiness, GitOps drift, security, capacity) score and triage a live cluster — every finding traces to a rule, not a model guess; the LLM only explains and prioritizes. Ships with a Helm chart, OpenTelemetry tracing, Prometheus/Grafana observability, and GitHub Actions CI.

RL Portfolio Optimization Platform | PyTorch, Stable-Baselines3, ONNX, FastAPI, Helm, ArgoCD

[GitHub](#)

- **SAC agent beat the S&P 500 by 5.6× out-of-sample** — 5 deep-RL agents (PPO, A2C, DDPG, SAC, TD3) and 4 classical optimizers (Markowitz, Min-Variance, Max-Sharpe, HRP) trained in Stable-Baselines3, exported to **ONNX**, served via **FastAPI** with per-day live inference. Production SRE layer: multi-burn-rate Prometheus SLOs, Helm/ArgoCD GitOps, and a full runbook.

distsched – Distributed Job Scheduler | Go, gRPC, Protobuf, BoltDB, Prometheus

[GitHub](#)

- **Fault-tolerant scheduler** whose correctness is proven by a chaos script that kills the leader under load and verifies clean failover; **Raft-style leader election** with term fencing, a priority queue, exponential-backoff retry with a dead-letter queue, DAG dependencies, and cron scheduling.

TECHNICAL SKILLS

Cloud & Infra: AWS (Lambda, S3, DynamoDB, API Gateway, IAM, CUR, EC2), IBM Cloud, Kubernetes, OpenShift/ROKS, Docker, Helm

DevOps & IaC: Terraform, Ansible, Jenkins, ArgoCD (GitOps), GitHub Actions

Observability & SRE: Prometheus, Grafana, Loki, Instana, Alertmanager, OpenTelemetry, SLO/SLI, Incident Mgmt, BCDR, RCA
AI / ML Systems: PyTorch, Stable-Baselines3, ONNX, Model Context Protocol (MCP), Multi-Agent Orchestration, RAG, Milvus, IBM Granite, TensorFlow

Languages: Python, Go, Java, C/C++, SQL, Bash, JavaScript/TypeScript

Databases: DynamoDB, IBM DB2, MySQL, MongoDB